

IT-Simplera Solutions



Week 2 Task Report IP Addressing (FLSM & VLSM) and VLAN Configuration

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Date Of Submission

8th July, 2026

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IP Addressing (FLSM & VLSM) and VLAN Configuration

Executive Summary

This technical report details the implementation of secure network segmentation and optimized IP address allocation as mandated for Week 02 of the network administration internship program. The primary focus of this phase is to eliminate address space waste within an enterprise framework using subnetting methodologies, while simultaneously establishing logical isolation at the data link layer.

The practical execution of these design methodologies is completed and verified using multi-platform environments across Cisco Packet Tracer and GNS3 emulation software. The foundational configurations established herein ensure a scalable, resilient, and highly secure infrastructure ready for enterprise-level deployment.

Internship Objectives

The deployment and configuration strategies implemented throughout this week focus on the following key operational milestones:

- **IP Engineering Mastery:** Applying Fixed Length Subnet Masking (FLSM) and Variable Length Subnet Masking (VLSM) mathematical frameworks to structure deterministic network boundaries.
- **Resource Optimization:** Eliminating IPv4 address exhaustion by aligning block allocations strictly with departmental host requirements.
- **Traffic Segmentation:** Restricting broadcast domains and isolating sensitive departmental data by implementing Virtual Local Area Networks (VLANs).
- **Multi-Platform Deployment:** Validating cross-environment configuration consistency by replicating topologies across Cisco Packet Tracer and GNS3 emulation platforms.
- **Enterprise Verification:** Using industry-standard Cisco IOS troubleshooting commands to verify path selection, trunk links, and packet transit success.

Core Technical Overview

FLSM vs. VLSM Paradigm

- **Fixed Length Subnet Masking (FLSM):** Divides a core network block into equal-sized subnets where every segment shares an identical subnet mask. While straightforward to administer, it introduces significant address space wastage when subnets host highly disparate node densities.

- **Variable Length Subnet Masking (VLSM):** Employs a hierarchical design approach where subnet masks vary in length. This allows subnets to match the exact host requirements of individual segments, conserving critical IP assets.

VLAN Infrastructure and Trunks

- **Virtual Local Area Networks (VLANs):** Break down a single physical broadcast domain into multiple logical broadcast domains at Layer 2. This ensures data privacy and limits broadcast storms across the infrastructure.
- **Access Ports:** Switchports configured to carry traffic for a single, specific VLAN destination.
- **Trunk Ports (IEEE 802.1Q):** High-capacity links configured between switches to carry traffic originating from multiple VLANs across a single physical medium.

Concise CIDR Reference Table (/32 to /22)

CIDR Prefix	Subnet Mask (Decimal)	Total IP Addresses	Usable Hosts
/32	255.255.255.255	1	0 (Host Route)
/31	255.255.255.254	2	2 (Point-to-Point links)
/30	255.255.255.252	4	2 (Standard Point-to-Point)
/29	255.255.255.248	8	6
/28	255.255.255.240	16	14
/27	255.255.255.224	32	30
/26	255.255.255.192	64	62
/25	255.255.255.128	128	126
/24	255.255.255.0	256	254

CIDR Prefix	Subnet Mask (Decimal)	Total IP Addresses	Usable Hosts
/23	255.255.254.0	512	510
/22	255.255.252.0	1,024	1,022

Fixed Length Subnet Masking

Definition of FLSM

Fixed Length Subnet Masking (FLSM) is an IPv4 addressing architecture where a single, uniform subnet mask is applied across all subnets derived from a base network. Under FLSM, every subnet is partitioned into identical block sizes, meaning each subnet contains an equal number of total IP addresses and supports the exact same maximum number of hosts, regardless of the actual host requirements of that network segment.

Important Mechanics & Structural Rules

- **Sizing to the Maximum Requirement:** When designing an FLSM network with varying host requirements, the subnet mask must be chosen based entirely on the **largest host requirement**. Every other subnet—even if it only requires a few IP addresses—is forced into that same large block size.
- **Constant Increment Size:** The boundaries between subnets increment by a static value (e.g., blocks of 32, 64, or 128), which makes sequential IP tracking straightforward but structurally rigid.
- **Formulaic Limitations:** Subnet boundaries must align with standard powers of 2 (2^n). The total number of subnets created is determined by 2^n (where n is the number of bits borrowed from the host portion), and the total IP capacity per subnet is 2^h (where h is the remaining host bits).

Critical Analysis: Core Characteristics

1. Low Computational Overhead

Because the subnet mask is constant across the entire administrative domain, routing updates carry minimal complexity. Traditional classful routing protocols (like RIPv1) natively support FLSM because they do not need to advertise subnet mask information along with network prefixes.

2. Severe IP Address Waste

The primary drawback of FLSM is inefficiency. If a network infrastructure requires one subnet for 60 hosts and three subnets for point-to-point router links (2 hosts each), FLSM forces all four subnets

to use a /26 mask (64 IPs per block). This leads to significant address exhaustion within the smaller subnets.

3. Simplified Management

Network addressing tables under FLSM are predictable and symmetric. Troubleshooting IP assignment overlaps is minimal because every subnet occupies an identical structural footprint.

Problems:

Problem 1:

Divide the network **192.168.10.0/24** into 4 equal subnets.

Solution for Problem 1: Divide into 4 subnets

1. Problem Scenario & Given Data

- Original Network Block: 192.168.10.0/24
- Requirement: Create 4 subnets of equal size

2. Step-by-Step Mathematical Calculations

Step 1: Determine the number of bits to borrow (n)

To find out how many host bits must be converted into network bits, use the subnet formula:

$$2^n \geq \text{Required Subnets}$$

Substitute the required subnets (4) into the formula:

$$2^n \geq 4 \Rightarrow n = 2 \text{ bits}$$

You must borrow 2 bits from the host portion.

Step 2: Determine the New Subnet Mask

- Original CIDR: /24 (Binary: 11111111.11111111.11111111.00000000)
- New CIDR: /24 + 2 = /26 (Binary: 11111111.11111111.11111111.11000000)

Convert the last octet from binary (11000000) to decimal:

$$128 + 64 = 192$$

- New Subnet Mask: 255.255.255.192

Step 3: Calculate the Total and Usable Hosts per Subnet

Count the remaining host bits (h). A standard Class C network has 8 host bits. Since 2 bits were borrowed, $8 - 2 = 6$ bits remain for hosts.

- Total IP Addresses per Subnet: $2^6 = 64$

- Usable Hosts per Subnet: $2^6 - 2 = 62$ (Subtracting 2 for the Network ID and Broadcast Address)

Step 4: Find the Magic Number (Block Size / Increment)

The magic number determines how the subnets advance. Subtract the custom octet value from 256:

$$256 - 192 = 64$$

The subnets will increment by 64 in the fourth octet.

3. Final Subnet Allocation Table

Subnet	Network ID	First Usable IP	Last Usable IP	Broadcast Address
Subnet 1	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

Problem 2:

Divide the network **192.168.50.0/24** into 8 equal subnets.

Solution for Problem 2: Divide into 8 subnets

1. Problem Scenario & Given Data

- Original Network Block: 192.168.50.0/24
- Requirement: Create at least 8 subnets to isolate small testing labs

2. Step-by-Step Mathematical Calculations

Step 1: Determine the number of bits to borrow (n)

To find the exact number of host bits to turn into network bits, use the subnet formula:

$$2^n \geq \text{Required Subnets}$$

Substitute the required number of subnets (8):

$$2^n \geq 8 \Rightarrow n = 3 \text{ bits}$$

You must borrow 3 bits from the host portion.

Step 2: Determine the New Subnet Mask

- Original CIDR: /24 (Binary: 11111111.11111111.11111111.00000000)
- New CIDR: /24 + 3 = /27 (Binary: 11111111.11111111.11111111.11100000)

Convert the last octet from binary (11100000) to decimal:

$$128 + 64 + 32 = 224$$

- New Subnet Mask: 255.255.255.224

Step 3: Calculate the Total and Usable Hosts per Subnet

Count the remaining host bits (h). A standard Class C network has 8 host bits. Since 3 bits were borrowed, $8 - 3 = 5$ bits remain for hosts.

- Total IP Addresses per Subnet: $2^5 = 32$
- Usable Hosts per Subnet: $2^5 - 2 = 30$ (Subtracting 2 for the Network ID and Broadcast Address)

Step 4: Find the Magic Number (Block Size / Increment)

Subtract the custom octet value from 256:

$$256 - 224 = 32$$

The subnets will increment by 32 in the fourth octet.

3. Subnet Allocation Table (First 4 Subnets)

Subnet	Network ID	First Usable IP	Last Usable IP	Broadcast Address
Subnet 1	192.168.50.0	192.168.50.1	192.168.50.30	192.168.50.31
Subnet 2	192.168.50.32	192.168.50.33	192.168.50.62	192.168.50.63
Subnet 3	192.168.50.64	192.168.50.65	192.168.50.94	192.168.50.95
Subnet 4	192.168.50.96	192.168.50.97	192.168.50.126	192.168.50.127

Problem 3:

Divide the network **192.168.1.0/24** into 2 equal subnets.

Solution for Problem 3: Divide into 2 subnets

1. Given Parameters

- Base Network: 192.168.1.0/24

- **Desired Subnets:** 2

2. Subnet Mask Calculation

- **Formula for Subnets:** $2^n \geq \text{Required Subnets}$ (where n is the number of borrowed network bits)
- **Calculation:** $2^1 = 2$
- **Bits Borrowed (n):** 1 bit
- **New Subnet Mask (CIDR):** /24 + 1 = /25
- **Subnet Mask in Binary:** 11111111.11111111.11111111.10000000
- **Custom Subnet Mask (Decimal):** 255.255.255.128

3. Host Calculations per Subnet

- **Remaining Host Bits (h):** $32 - 25 = 7$ bits
- **Total IP Addresses per Subnet:** $2^7 = 128$
- **Usable Hosts per Subnet:** $2^7 - 2 = 126$
- **Block Size (Increment):** $256 - 128 = 128$

4. Subnet Allocation Table

Subnet #	Network Address	Usable Host IP Range	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1 to 192.168.1.126	192.168.1.127
Subnet 2	192.168.1.128	192.168.1.129 to 192.168.1.254	192.168.1.255

Problem 4:

Divide the network **192.168.1.0/24** into 3 equal subnets.

Solution for Problem 4: Divide into 3 Subnets

1. Given Parameters

- **Base Network:** 192.168.1.0/24
- **Desired Subnets:** 3

2. Subnet Mask Calculation

- **Formula for Subnets:** $2^n \geq \text{Required Subnets}$ (where n is the number of borrowed network bits)

- **Calculation:** $2^2 = 4$ (We must borrow 2 bits, which creates a maximum of 4 subnets, to satisfy the requirement of 3 subnets)
- **Bits Borrowed (n):** 2 bits
- **New Subnet Mask (CIDR):** $/24 + 2 = /26$
- **Subnet Mask in Binary:** 11111111.11111111.11111111.11000000
- **Custom Subnet Mask (Decimal):** 255.255.255.192

3. Host Calculations per Subnet

- **Remaining Host Bits (h):** $32 - 26 = 6$ bits
- **Total IP Addresses per Subnet:** $2^6 = 64$
- **Usable Hosts per Subnet:** $2^6 - 2 = 62$
- **Block Size (Increment):** $256 - 192 = 64$

4. Subnet Allocation Table

(Note: This configuration yields 4 total subnets; the first 3 fulfill the requirement)

Subnet #	Network Address	Usable Host IP Range	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1 to 192.168.1.62	192.168.1.63
Subnet 2	192.168.1.64	192.168.1.65 to 192.168.1.126	192.168.1.127
Subnet 3	192.168.1.128	192.168.1.129 to 192.168.1.190	192.168.1.191

Problem 5:

Divide the network **192.168.10.0/24** into 16 equal subnets.

Solution for Problem 5: Dividing into 16 subnets

1. Given Parameters

- **Base Network:** 192.168.10.0/24
- **Desired Subnets:** 16

2. Subnet Mask Calculation

- **Formula for Subnets:** $2^n \geq \text{Required Subnets}$ (where n is the number of borrowed network bits)

- **Calculation:** $2^4 = 16$
- **Bits Borrowed (n):** 4 bits
- **New Subnet Mask (CIDR):** $/24 + 4 = /28$
- **Subnet Mask in Binary:** 11111111.11111111.11111111.11110000
- **Custom Subnet Mask (Decimal):** 255.255.255.240

3. Host Calculations per Subnet

- **Remaining Host Bits (h):** $32 - 28 = 4$ bits
- **Total IP Addresses per Subnet:** $2^4 = 16$
- **Usable Hosts per Subnet:** $2^4 - 2 = 14$
- **Block Size (Increment):** $256 - 240 = 16$

4. Subnet Allocation Table

Subnet #	Network Address	Usable Host IP Range	Broadcast Address
Subnet 1	192.168.10.0	192.168.10.1 to 192.168.10.14	192.168.10.15
Subnet 2	192.168.10.16	192.168.10.17 to 192.168.10.30	192.168.10.31
Subnet 3	192.168.10.32	192.168.10.33 to 192.168.10.46	192.168.10.47
Subnet 4	192.168.10.48	192.168.10.49 to 192.168.10.62	192.168.10.63
Subnet 5	192.168.10.64	192.168.10.65 to 192.168.10.78	192.168.10.79
Subnet 6	192.168.10.80	192.168.10.81 to 192.168.10.94	192.168.10.95

Problem 6:

Divide the network **192.168.10.0/24** into 32 equal subnets.

Solution for Problem 6: Divide into 32 Subnets

1. Given Parameters

- **Base Network:** 192.168.10.0/24
- **Desired Subnets:** 32

2. Subnet Mask Calculation

- **Formula for Subnets:** $2^n \geq \text{Required Subnets}$ (where n is the number of borrowed network bits)
- **Calculation:** $2^5 = 32$
- **Bits Borrowed (n):** 5 bits
- **New Subnet Mask (CIDR):** $/24 + 5 = /29$
- **Subnet Mask in Binary:** 11111111.11111111.11111111.11111000
- **Custom Subnet Mask (Decimal):** 255.255.255.248

3. Host Calculations per Subnet

- **Remaining Host Bits (h):** $32 - 29 = 3$ bits
- **Total IP Addresses per Subnet:** $2^3 = 8$
- **Usable Hosts per Subnet:** $2^3 - 2 = 6$
- **Block Size (Increment):** $256 - 248 = 8$

4. Subnet Allocation Table

Subnet #	Network Address	Usable Host IP Range	Broadcast Address
Subnet 1	192.168.10.0	192.168.10.1 to 192.168.10.6	192.168.10.7
Subnet 2	192.168.10.8	192.168.10.9 to 192.168.10.14	192.168.10.15
Subnet 3	192.168.10.16	192.168.10.17 to 192.168.10.22	192.168.10.23
Subnet 4	192.168.10.24	192.168.10.25 to 192.168.10.30	192.168.10.31
Subnet 31	192.168.10.240	192.168.10.241 to 192.168.10.246	192.168.10.247
Subnet 32	192.168.10.248	192.168.10.249 to 192.168.10.254	192.168.10.255

Problem 7:

Divide the network **200.10.1.0/24** into 64 equal subnets.

Solution for Problem 7: Divide into 64 Subnets

1. Given Parameters

- **Base Network:** 200.10.1.0/24
- **Desired Subnets:** 64

2. Subnet Mask Calculation

- **Formula for Subnets:** $2^n \geq \text{Required Subnets}$ (where n is the number of borrowed network bits)
- **Calculation:** $2^6 = 64$
- **Bits Borrowed (n):** 6 bits
- **New Subnet Mask (CIDR):** /24 + 6 = /30
- **Subnet Mask in Binary:** 11111111.11111111.11111111.11111100
- **Custom Subnet Mask (Decimal):** 255.255.255.252

3. Host Calculations per Subnet

- **Remaining Host Bits (h):** $32 - 30 = 2$ bits
- **Total IP Addresses per Subnet:** $2^2 = 4$
- **Usable Hosts per Subnet:** $2^2 - 2 = 2$
- **Block Size (Increment):** $256 - 252 = 4$

4. Subnet Allocation Table (First 5 & Last 5 Subnets)

Subnet #	Network Address	Usable Host IP Range	Broadcast Address
Subnet 1	200.10.1.0	200.10.1.1 to 200.10.1.2	200.10.1.3
Subnet 2	200.10.1.4	200.10.1.5 to 200.10.1.6	200.10.1.7
Subnet 3	200.10.1.8	200.10.1.9 to 200.10.1.10	200.10.1.11
Subnet 4	200.10.1.12	200.10.1.13 to 200.10.1.14	200.10.1.15
Subnet 5	200.10.1.16	200.10.1.17 to 200.10.1.18	200.10.1.19
Subnet 63	200.10.1.248	200.10.1.249 to 200.10.1.250	200.10.1.251

Subnet #	Network Address	Usable Host IP Range	Broadcast Address
Subnet 64	200.10.1.252	200.10.1.253 to 200.10.1.254	200.10.1.255

Problem 8:

Divide the network **198.168.10.0/24** into 4 subnets. Where subnet 1 has 30 hosts, subnet 2 has 15 hosts, subnet 3 has 20 hosts and subnet 4 has 5 hosts. Also calculate the used IP's and waste IP's.

Problem 8 Solution: Fixed Length Subnet Masking (FLSM)

If you must apply **FLSM** instead of VLSM for this problem, all subnets must be of equal size. To choose the block size, we must design for the largest requirement, which is **Subnet 1 (30 hosts)**.

1. Subnet Calculations

Step 1: Determine bits required for the largest subnet

- **Requirement:** 30 hosts
- **Formula:** $2^h - 2 \geq 30 \Rightarrow 2^5 - 2 = 30$ ($h = 5$ host bits needed)

Step 2: Calculate the Subnet Mask

- **Bits Borrowed (n):** $32 - 5 = 27$ bits total $\Rightarrow$ /27 network mask.
- **Custom Subnet Mask (Decimal):** 255.255.255.224
- **Block Size (Increment):** $2^5 = 32$

2. Subnet Allocation Table

Using a uniform block size of 32 for all 4 subnets:

Subnet Name	Assigned Hosts	Network Address	Usable Host IP Range	Broadcast Address	Total Allocated IPs
Subnet 1	30	198.168.10.0 /27	198.168.10.1 to 198.168.10.30	198.168.10.31	32
Subnet 2	15	198.168.10.32 /27	198.168.10.33 to 198.168.10.62	198.168.10.63	32

Subnet Name	Assigned Hosts	Network Address	Usable Host IP Range	Broadcast Address	Total Allocated IPs
Subnet 3	20	198.168.10.64 /27	198.168.10.65 to 198.168.10.94	198.168.10.95	32
Subnet 4	5	198.168.10.96 /27	198.168.10.97 to 198.168.10.126	198.168.10.127	32

3. IP Usage and Waste Analysis

- **Total IPs Allocated in Blocks:** $32 \times 4 = 128$ IPs
- **Used IPs (Requested Hosts):** $30 + 15 + 20 + 5 = 70$ IPs
- **Wasted/Unused IPs inside subnets:** Total Allocated – Used = $128 - 70 = 58$ IPs

Breakdown of Wasted IPs under FLSM:

- **Network & Broadcast overhead:** $2 \text{ IPs} \times 4 \text{ subnets} = 8 \text{ IPs}$
- **Unassigned Usable IPs inside blocks due to fixed sizing:**
 - Subnet 1: $30 - 30 = 0$ IPs
 - Subnet 2: $30 - 15 = 15$ IPs
 - Subnet 3: $30 - 20 = 10$ IPs
 - Subnet 4: $30 - 5 = 25$ IPs
- **Total Waste:** $8 + 0 + 15 + 10 + 25 = 58$ IPs

Problem 9:

Divide the network **198.168.10.0/24** into 3 subnets. Where subnet 1 has 50 hosts, subnet 2 has 21 hosts, subnet 3 has 32 hosts. Also calculate the used IP's and waste IP's.

Problem 9 Solution: Fixed Length Subnet Masking (FLSM)

To solve this problem using **FLSM**, all subnets must be configured with a uniform block size. The size of the subnets is determined by the largest host requirement in the network, which is **Subnet 1 (50 hosts)**.

1. Subnet Calculations

Step 1: Determine bits required for the largest subnet

- **Requirement:** 50 hosts
- **Formula:** $2^h - 2 \geq \text{Hosts Requirement}$
- **Calculation:** $2^6 - 2 = 62$ hosts (Using $h = 6$ host bits provides enough addresses, whereas $2^5 - 2 = 30$ is too small)
- **Host bits (h):** 6 bits

Step 2: Calculate the Subnet Mask

- **Bits Borrowed (n):** $32 - 6 = 26$ bits total $\Rightarrow$ /26 network mask.
- **Custom Subnet Mask (Decimal):** 255.255.255.192
- **Block Size (Increment):** $2^6 = 64$

2. Subnet Allocation Table

Using a uniform block size of 64 for all 3 required subnets:

Subnet Name	Required Hosts	Network Address	Usable Host IP Range	Broadcast Address	Total Allocated IPs
Subnet 1	50	198.168.10.0 /26	198.168.10.1 to 198.168.10.62	198.168.10.63	64
Subnet 2	21	198.168.10.64 /26	198.168.10.65 to 198.168.10.126	198.168.10.127	64
Subnet 3	32	198.168.10.128 /26	198.168.10.129 to 198.168.10.190	198.168.10.191	64

3. IP Usage and Waste Analysis

- **Total IPs Allocated across the 3 Blocks:** $64 \times 3 = 192$ IPs
- **Used IPs (Requested Hosts):** $50 + 21 + 32 = 103$ IPs
- **Wasted/Unused IPs inside subnets:** Total Allocated – Used = $192 - 103 = 89$ IPs

Breakdown of Wasted IPs under FLSM:

- **Network & Broadcast overhead:** $2 \text{ IPs} \times 3 \text{ subnets} = 6 \text{ IPs}$

- **Unassigned Usable IPs inside blocks due to fixed sizing:**
 - Subnet 1: $62 - 50 = 12$ IPs
 - Subnet 2: $62 - 21 = 41$ IPs
 - Subnet 3: $62 - 32 = 30$ IPs
- **Total Waste:** $6 + 12 + 41 + 30 = 89$ IPs

Problem 10:

Divide the network **198.168.8.0/22** into 2 subnets. Each subnet has 300 hosts each.

Solution for Problem 10: Divide into 2 subnets with 300 hosts

1. Given Parameters

- **New Base Network:** 192.168.8.0/22
- **Desired Subnets:** 2
- **Hosts per Subnet:** 300

2. Subnet Mask Calculation

- **Formula for Subnets:** $2^n \geq 2 \Rightarrow n = 1$ bit borrowed.
- **New Subnet Mask (CIDR):** /22 + 1 = /23
- **Custom Subnet Mask (Decimal):** 255.255.254.0

3. Host Calculations per Subnet

- **Remaining Host Bits (h):** $32 - 23 = 9$ bits
- **Total IP Addresses per Subnet:** $2^9 = 512$
- **Usable Hosts per Subnet:** $512 - 2 = 510$ (This safely accommodates your 300 hosts)
- **Block Size (Increment):** 2 in the third octet.

4. Adjusted Subnet Allocation Table

Subnet #	Network Address	Usable Host IP Range	Broadcast Address
Subnet 1	192.168.8.0/23	192.168.8.1 to 192.168.9.254	192.168.9.255
Subnet 2	192.168.10.0/23	192.168.10.1 to 192.168.11.254	192.168.11.255

Variable Length Subnet Masking

Definition of VLSM

Variable Length Subnet Masking (VLSM) is an advanced IPv4 addressing design method that allows a network administrator to use different subnet masks for different subnets derived from a single base network. Instead of dividing a network into equal-sized blocks (as in FLSM), VLSM allows you to subnet a subnet, matching the block size precisely to the specific host requirements of each individual network segment.

Core Mechanics & Structural Rules

- **Largest-to-Smallest Ordering:** To prevent address overlapping, VLSM requires sorting host requirements from the largest to the smallest before calculating masks.
- **Subnetting a Subnet:** VLSM breaks a single block down into smaller portions recursively. For example, a /24 block can be split into two /25 blocks, and one of those /25 blocks can be split further into two /26 blocks.
- **Classless Routing Dependency:** Because subnet masks vary throughout the topology, VLSM requires classless routing protocols (such as OSPF, EIGRP, or RIPv2) that explicitly carry the subnet mask info in their routing updates.

Key Advantages of VLSM

1. Maximum IP Efficiency

VLSM minimizes address waste by tailoring block sizes. Point-to-point serial links can be assigned a /30 or /31 mask (allocating only 2 usable hosts), while large departments can use a /26 mask (allocating 62 usable hosts), avoiding the massive footprint waste seen in FLSM.

2. Hierarchical Address Summarization

VLSM allows for clean route summarization (aggregation). Multiple small subnets can be represented as a single summary route in the upstream routing tables, drastically reducing router CPU cycles and overhead.

3. Flexible Network Architecture

Administrators can easily allocate variable block sizes for future expansion without restructuring the entire IP plan or modifying unaffected subnets.

Scenarios

Scenario 1:

Base Network IP: 192.168.1.0/24

Subnet Requirements:

- Subnet A: 49 hosts

- Subnet B: 25 hosts
- Subnet C: 14 hosts
- Subnet D: 6 hosts

Scenario 1 VLSM Step-by-Step Calculations

1. Sort Requirements (Largest to Smallest)

1. **Subnet A:** 49 hosts
2. **Subnet B:** 25 hosts
3. **Subnet C:** 14 hosts
4. **Subnet D:** 6 hosts

2. Step-by-Step Subnet Calculations

Subnet A: 49 Hosts

- **Host Bits (h):** $2^h - 2 \geq 49 \Rightarrow 2^6 - 2 = 62 \Rightarrow h = 6$
- **New Mask (CIDR):** $32 - 6 = /26$
- **Custom Subnet Mask (Decimal):** 255.255.255.192
- **Total IP Addresses Allocated:** $2^6 = 64$
- **Network Address:** 192.168.1.0
- **Usable Host IP Range:** 192.168.1.1 to 192.168.1.62
- **Broadcast Address:** 192.168.1.63

Subnet B: 25 Hosts

- **Host Bits (h):** $2^h - 2 \geq 25 \Rightarrow 2^5 - 2 = 30 \Rightarrow h = 5$
- **New Mask (CIDR):** $32 - 5 = /27$
- **Custom Subnet Mask (Decimal):** 255.255.255.224
- **Total IP Addresses Allocated:** $2^5 = 32$
- **Network Address:** 192.168.1.64
- **Usable Host IP Range:** 192.168.1.65 to 192.168.1.94
- **Broadcast Address:** 192.168.1.95

Subnet C: 14 Hosts

- **Host Bits (h):** $2^h - 2 \geq 14 \Rightarrow 2^4 - 2 = 14 \Rightarrow h = 4$
- **New Mask (CIDR):** $32 - 4 = /28$
- **Custom Subnet Mask (Decimal):** 255.255.255.240

- **Total IP Addresses Allocated:** $2^4 = 16$
- **Network Address:** 192.168.1.96
- **Usable Host IP Range:** 192.168.1.97 to 192.168.1.110
- **Broadcast Address:** 192.168.1.111

Subnet D: 6 Hosts

- **Host Bits (h):** $2^h - 2 \geq 6 \Rightarrow 2^3 - 2 = 6 \Rightarrow h = 3$
- **New Mask (CIDR):** $32 - 3 = /29$
- **Custom Subnet Mask (Decimal):** 255.255.255.248
- **Total IP Addresses Allocated:** $2^3 = 8$
- **Network Address:** 192.168.1.112
- **Usable Host IP Range:** 192.168.1.113 to 192.168.1.118
- **Broadcast Address:** 192.168.1.119

3. Subnet Allocation Table

Subnet Name	Required Hosts	Network Address	Subnet Mask (Decimal)	Usable Host IP Range	Broadcast Address	Total Allocated IPs
Subnet A	49	192.168.1.0 /26	255.255.255.192	192.168.1.1 to 192.168.1.62	192.168.1.63	64
Subnet B	25	192.168.1.64 /27	255.255.255.224	192.168.1.65 to 192.168.1.94	192.168.1.95	32
Subnet C	14	192.168.1.96 /28	255.255.255.240	192.168.1.97 to 192.168.1.110	192.168.1.111	16
Subnet D	6	192.168.1.112 /29	255.255.255.248	192.168.1.113 to 192.168.1.118	192.168.1.119	8

4. IP Usage and Waste Analysis

- **Total IPs Allocated:** $64 + 32 + 16 + 8 = 120$ IPs
- **Used IPs (Requested Hosts):** $49 + 25 + 14 + 6 = 94$ IPs
- **Wasted/Unused IPs inside subnets:** $120 - 94 = 26$ IPs
 - *Network & Broadcast Overhead:* $2 \text{ IPs} \times 4 \text{ subnets} = 8 \text{ IPs}$

- *Unassigned Usable IPs inside blocks:* 18 IPs (13 in Subnet A, 5 in Subnet B, 0 in Subnet C, 0 in Subnet D)

Scenario 2:

Base Network IP: 192.168.1.0/24

University has 3 departments in one building. Assign IP's using VLSM for each department according to the host requirements.

Subnet Requirements:

- Telecom: 29 hosts
- IT: 12 hosts
- AI: 54 hosts

Scenario 2 VLSM Step-by-Step Calculations

1. Sort Requirements (Largest to Smallest)

1. **AI Department:** 54 hosts
2. **Telecom Department:** 29 hosts
3. **IT Department:** 12 hosts

2. Step-by-Step Subnet Calculations

AI Department: 54 Hosts

- **Host Bits (h):** $2^h - 2 \geq 54 \Rightarrow 2^6 - 2 = 62 \Rightarrow h = 6$
- **New Mask (CIDR):** $32 - 6 = /26$
- **Custom Subnet Mask (Decimal):** 255.255.255.192
- **Total IP Addresses Allocated:** $2^6 = 64$
- **Network Address:** 192.168.1.0
- **Usable Host IP Range:** 192.168.1.1 to 192.168.1.62
- **Broadcast Address:** 192.168.1.63

Telecom Department: 29 Hosts

- **Host Bits (h):** $2^h - 2 \geq 29 \Rightarrow 2^5 - 2 = 30 \Rightarrow h = 5$
- **New Mask (CIDR):** $32 - 5 = /27$
- **Custom Subnet Mask (Decimal):** 255.255.255.224
- **Total IP Addresses Allocated:** $2^5 = 32$
- **Network Address:** 192.168.1.64

- **Usable Host IP Range:** 192.168.1.65 to 192.168.1.94
- **Broadcast Address:** 192.168.1.95

IT Department: 12 Hosts

- **Host Bits (h):** $2^h - 2 \geq 12 \Rightarrow 2^4 - 2 = 14 \Rightarrow h = 4$
- **New Mask (CIDR):** $32 - 4 = /28$
- **Custom Subnet Mask (Decimal):** 255.255.255.240
- **Total IP Addresses Allocated:** $2^4 = 16$
- **Network Address:** 192.168.1.96
- **Usable Host IP Range:** 192.168.1.97 to 192.168.1.110
- **Broadcast Address:** 192.168.1.111

3. Subnet Allocation Table

Subnet Name	Required Hosts	Network Address	Subnet Mask (Decimal)	Usable Host IP Range	Broadcast Address	Total Allocated IPs
AI	54	192.168.1.0 /26	255.255.255.192	192.168.1.1 to 192.168.1.62	192.168.1.63	64
Telecom	29	192.168.1.64 /27	255.255.255.224	192.168.1.65 to 192.168.1.94	192.168.1.95	32
IT	12	192.168.1.96 /28	255.255.255.240	192.168.1.97 to 192.168.1.110	192.168.1.111	16

4. IP Usage and Waste Analysis

- **Total IPs Allocated:** $64 + 32 + 16 = 112$ IPs
- **Used IPs (Requested Hosts):** $54 + 29 + 12 = 95$ IPs
- **Wasted/Unused IPs inside subnets:** $112 - 95 = 17$ IPs
 - *Network & Broadcast Overhead:* $2 \text{ IPs} \times 3 \text{ subnets} = 6 \text{ IPs}$
 - *Unassigned Usable IPs inside blocks:* 11 IPs (8 in AI, 1 in Telecom, 2 in IT)

Scenario 3:

Base Network IP: 200.10.0.0/23

Department of Telecom has divided its network into 3 parts for different users.

Subnet Requirements:

- Management: 62 hosts
- Guests: 30 hosts
- Students: 350 hosts

Scenario 3 VLSM Step-by-Step Calculations

1. Sort Requirements (Largest to Smallest)

1. **Students:** 350 hosts
2. **Management:** 62 hosts
3. **Guests:** 30 hosts

2. Step-by-Step Subnet Calculations

Students: 350 Hosts

- **Host Bits (h):** $2^h - 2 \geq 350 \Rightarrow 2^9 - 2 = 510 \Rightarrow h = 9$
- **New Mask (CIDR):** $32 - 9 = /23$
- **Custom Subnet Mask (Decimal):** 255.255.254.0
- **Total IP Addresses Allocated:** $2^9 = 512$
- **Network Address:** 200.10.0.0
- **Usable Host IP Range:** 200.10.0.1 to 200.10.1.254
- **Broadcast Address:** 200.10.1.255

Management: 62 Hosts

- **Host Bits (h):** $2^h - 2 \geq 62 \Rightarrow 2^6 - 2 = 62 \Rightarrow h = 6$
- **New Mask (CIDR):** $32 - 6 = /26$
- **Custom Subnet Mask (Decimal):** 255.255.255.192
- **Total IP Addresses Allocated:** $2^6 = 64$
- **Network Address:** 200.10.2.0
- **Usable Host IP Range:** 200.10.2.1 to 200.10.2.62
- **Broadcast Address:** 200.10.2.63

Guests: 30 Hosts

- **Host Bits (h):** $2^h - 2 \geq 30 \Rightarrow 2^5 - 2 = 30 \Rightarrow h = 5$
- **New Mask (CIDR):** $32 - 5 = /27$
- **Custom Subnet Mask (Decimal):** 255.255.255.224
- **Total IP Addresses Allocated:** $2^5 = 32$
- **Network Address:** 200.10.2.64

- **Usable Host IP Range:** 200.10.2.65 to 200.10.2.94
- **Broadcast Address:** 200.10.2.95

3. Subnet Allocation Table

Subnet Name	Required Hosts	Network Address	Subnet Mask (Decimal)	Usable Host IP Range	Broadcast Address	Total Allocated IPs
Students	350	200.10.0.0 /23	255.255.248.0	200.10.0.1 to 200.10.1.254	200.10.1.255	512
Management	62	200.10.2.0 /26	255.255.255.192	200.10.2.1 to 200.10.2.62	200.10.2.63	64
Guests	30	200.10.2.64 /27	255.255.255.224	200.10.2.65 to 200.10.2.94	200.10.2.95	32

4. IP Usage and Waste Analysis

- **Total IPs Allocated:** $512 + 64 + 32 = 608$ IPs
- **Used IPs (Requested Hosts):** $350 + 62 + 30 = 442$ IPs
- **Wasted/Unused IPs inside subnets:** $608 - 442 = 166$ IPs
 - *Network & Broadcast Overhead:* $2 \text{ IPs} \times 3 \text{ subnets} = 6 \text{ IPs}$
 - *Unassigned Usable IPs inside blocks:* 160 IPs (160 in Students, 0 in Management, 0 in Guests)

Scenario 4:

Base Network IP: 200.10.0.0/24

A grocery store has three sectors. Assign IP's according to each sector requirement.

Subnet Requirements:

- Management: 10 hosts
- Guests: 55 hosts
- Sales: 20 hosts

Scenario 4 VLSM Step-by-Step Calculations

1. Sort Requirements (Largest to Smallest)

1. **Guests:** 55 hosts
2. **Sales:** 20 hosts
3. **Management:** 10 hosts

2. Step-by-Step Subnet Calculations

Guests: 55 Hosts

- **Host Bits (h):** $2^h - 2 \geq 55 \Rightarrow 2^6 - 2 = 62 \Rightarrow h = 6$
- **New Mask (CIDR):** $32 - 6 = /26$
- **Custom Subnet Mask (Decimal):** 255.255.255.192
- **Total IP Addresses Allocated:** $2^6 = 64$
- **Network Address:** 200.10.0.0
- **Usable Host IP Range:** 200.10.0.1 to 200.10.0.62
- **Broadcast Address:** 200.10.0.63

Sales: 20 Hosts

- **Host Bits (h):** $2^h - 2 \geq 20 \Rightarrow 2^5 - 2 = 30 \Rightarrow h = 5$
- **New Mask (CIDR):** $32 - 5 = /27$
- **Custom Subnet Mask (Decimal):** 255.255.255.224
- **Total IP Addresses Allocated:** $2^5 = 32$
- **Network Address:** 200.10.0.64
- **Usable Host IP Range:** 200.10.0.65 to 200.10.0.94
- **Broadcast Address:** 200.10.0.95

Management: 10 Hosts

- **Host Bits (h):** $2^h - 2 \geq 10 \Rightarrow 2^4 - 2 = 14 \Rightarrow h = 4$
- **New Mask (CIDR):** $32 - 4 = /28$
- **Custom Subnet Mask (Decimal):** 255.255.255.240
- **Total IP Addresses Allocated:** $2^4 = 16$
- **Network Address:** 200.10.0.96
- **Usable Host IP Range:** 200.10.0.97 to 200.10.0.110
- **Broadcast Address:** 200.10.0.111

3. Subnet Allocation Table

Subnet Name	Required Hosts	Network Address	Subnet Mask (Decimal)	Usable Host IP Range	Broadcast Address	Total Allocated IPs
Guests	55	200.10.0.0 /26	255.255.255.192	200.10.0.1 to 200.10.0.62	200.10.0.63	64

Subnet Name	Required Hosts	Network Address	Subnet Mask (Decimal)	Usable Host IP Range	Broadcast Address	Total Allocated IPs
Sales	20	200.10.0.64 /27	255.255.255.224	200.10.0.65 to 200.10.0.94	200.10.0.95	32
Management	10	200.10.0.96 /28	255.255.255.240	200.10.0.97 to 200.10.0.110	200.10.0.111	16

4. IP Usage and Waste Analysis

- **Total IPs Allocated:** $64 + 32 + 16 = 112$ IPs
- **Used IPs (Requested Hosts):** $55 + 20 + 10 = 85$ IPs
- **Wasted/Unused IPs inside subnets:** $112 - 85 = 27$ IPs
 - *Network & Broadcast Overhead:* $2 \text{ IPs} \times 3 \text{ subnets} = 6 \text{ IPs}$
 - *Unassigned Usable IPs inside blocks:* 21 IPs (7 in Guests, 10 in Sales, 4 in Management)

Scenario 5:

Base Network IP: 192.168.10.0/24

Subnet Requirements:

- Subnet A: 78 hosts
- Subnet B: 55 hosts
- Subnet C: 13 hosts
- Subnet D: 5 hosts
- Subnet E: 1 host



Scenario 5 VLSM Step-by-Step Calculations

1. Sort Requirements (Largest to Smallest)

1. **Subnet A:** 78 hosts
2. **Subnet B:** 55 hosts
3. **Subnet C:** 13 hosts
4. **Subnet D:** 5 hosts
5. **Subnet E:** 1 host

2. Step-by-Step Subnet Calculations

Subnet A: 78 Hosts

- **Host Bits (h):** $2^h - 2 \geq 78 \Rightarrow 2^7 - 2 = 126 \Rightarrow h = 7$
- **New Mask (CIDR):** $32 - 7 = /25$
- **Custom Subnet Mask (Decimal):** 255.255.255.128
- **Total IP Addresses Allocated:** $2^7 = 128$
- **Network Address:** 192.168.10.0
- **Usable Host IP Range:** 192.168.10.1 to 192.168.10.126
- **Broadcast Address:** 192.168.10.127

Subnet B: 55 Hosts

- **Host Bits (h):** $2^h - 2 \geq 55 \Rightarrow 2^6 - 2 = 62 \Rightarrow h = 6$
- **New Mask (CIDR):** $32 - 6 = /26$
- **Custom Subnet Mask (Decimal):** 255.255.255.192
- **Total IP Addresses Allocated:** $2^6 = 64$
- **Network Address:** 192.168.10.128
- **Usable Host IP Range:** 192.168.10.129 to 192.168.10.190
- **Broadcast Address:** 192.168.10.191

Subnet C: 13 Hosts

- **Host Bits (h):** $2^h - 2 \geq 13 \Rightarrow 2^4 - 2 = 14 \Rightarrow h = 4$
- **New Mask (CIDR):** $32 - 4 = /28$
- **Custom Subnet Mask (Decimal):** 255.255.255.240
- **Total IP Addresses Allocated:** $2^4 = 16$
- **Network Address:** 192.168.10.192
- **Usable Host IP Range:** 192.168.10.193 to 192.168.10.206
- **Broadcast Address:** 192.168.10.207

Subnet D: 5 Hosts

- **Host Bits (h):** $2^h - 2 \geq 5 \Rightarrow 2^3 - 2 = 6 \Rightarrow h = 3$
- **New Mask (CIDR):** $32 - 3 = /29$
- **Custom Subnet Mask (Decimal):** 255.255.255.248
- **Total IP Addresses Allocated:** $2^3 = 8$
- **Network Address:** 192.168.10.208
- **Usable Host IP Range:** 192.168.10.209 to 192.168.10.214
- **Broadcast Address:** 192.168.10.215

Subnet E: 1 Host

- **Host Bits (h):** $2^h - 2 \geq 1 \Rightarrow 2^2 - 2 = 2 \Rightarrow h = 2$
- **New Mask (CIDR):** $32 - 2 = /30$
- **Custom Subnet Mask (Decimal):** 255.255.255.252
- **Total IP Addresses Allocated:** $2^2 = 4$
- **Network Address:** 192.168.10.216
- **Usable Host IP Range:** 192.168.10.217 to 192.168.10.218
- **Broadcast Address:** 192.168.10.219

3. Subnet Allocation Table

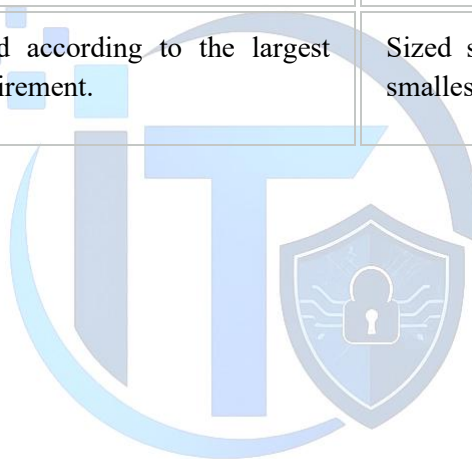
Subnet Name	Required Hosts	Network Address	Subnet Mask (Decimal)	Usable Host IP Range	Broadcast Address	Total Allocated IPs
Subnet A	78	192.168.10.0 /25	255.255.255.128	192.168.10.1 to 192.168.10.126	192.168.10.127	128
Subnet B	55	192.168.10.128 /26	255.255.255.192	192.168.10.129 to 192.168.10.190	192.168.10.191	64
Subnet C	13	192.168.10.192 /28	255.255.255.240	192.168.10.193 to 192.168.10.206	192.168.10.207	16
Subnet D	5	192.168.10.208 /29	255.255.255.248	192.168.10.209 to 192.168.10.214	192.168.10.215	8
Subnet E	1	192.168.10.216 /30	255.255.255.252	192.168.10.217 to 192.168.10.218	192.168.10.219	4

4. IP Usage and Waste Analysis

- **Total IPs Allocated:** $128 + 64 + 16 + 8 + 4 = 220$ IPs
- **Used IPs (Requested Hosts):** $78 + 55 + 13 + 5 + 1 = 152$ IPs
- **Wasted/Unused IPs inside subnets:** $220 - 152 = 68$ IPs
 - *Network & Broadcast Overhead:* $2 \text{ IPs} \times 5 \text{ subnets} = 10 \text{ IPs}$
 - *Unassigned Usable IPs inside blocks:* 58 IPs (48 in Subnet A, 7 in Subnet B, 1 in Subnet C, 1 in Subnet D, 1 in Subnet E)

FLSM vs. VLSM Structural Comparison

Feature	Fixed Length Subnet Masking (FLSM)	Variable Length Subnet Masking (VLSM)
Subnet Mask Size	Uniform. Single mask used globally across all subnets.	Variable. Custom mask optimized per individual subnet.
Subnet Block Sizes	Identical sizes for every partition.	Unique sizes customized to fit host requirements.
IP Efficiency	Low; high amounts of wasted or unused IPs.	High; minimizes waste by packing allocations tightly.
Routing Protocol Support	Works with both Classful (RIPv1) and Classless.	Requires Classless routing protocols (OSPF, EIGRP, RIPv2).
Design Sequence	Sized according to the largest requirement.	Sized sequentially from largest to smallest requirement.



VLAN Configuration

Objective

The goal of this task is to divide one physical network into three separate VLANs. This keeps traffic for each department isolated and improves security and organization on the network. Two switches are connected with a trunk link so devices in the same VLAN can communicate with each other even when they sit on different switches.

VLAN Plan

Three VLANs were created, one for each department. Each VLAN uses its own subnet.

VLAN ID	Name	Subnet	Assigned Devices
10	STAFF	192.168.1.0/24	PC1, PC2 (Switch1) / PC3, PC4 (Switch2)
20	SALES	192.168.2.0/24	PC5, PC6 (Switch1) / PC7, PC8 (Switch2)
30	HR	192.168.3.0/24	PC9, PC10 (Switch1) / PC11, PC12 (Switch2)

Port Assignment Plan

Port Range	Mode	VLAN
Fa0/1 - Fa0/4	Access	10 (STAFF)
Fa0/5 - Fa0/8	Access	20 (SALES)
Fa0/9 - Fa0/12	Access	30 (HR)
Fa0/24	Trunk (802.1Q)	1, 10, 20, 30

Network Topology

The topology below shows Switch1 and Switch2 connected by a trunk link. Each switch has four PCs split across the three VLANs, with matching VLANs on both switches.

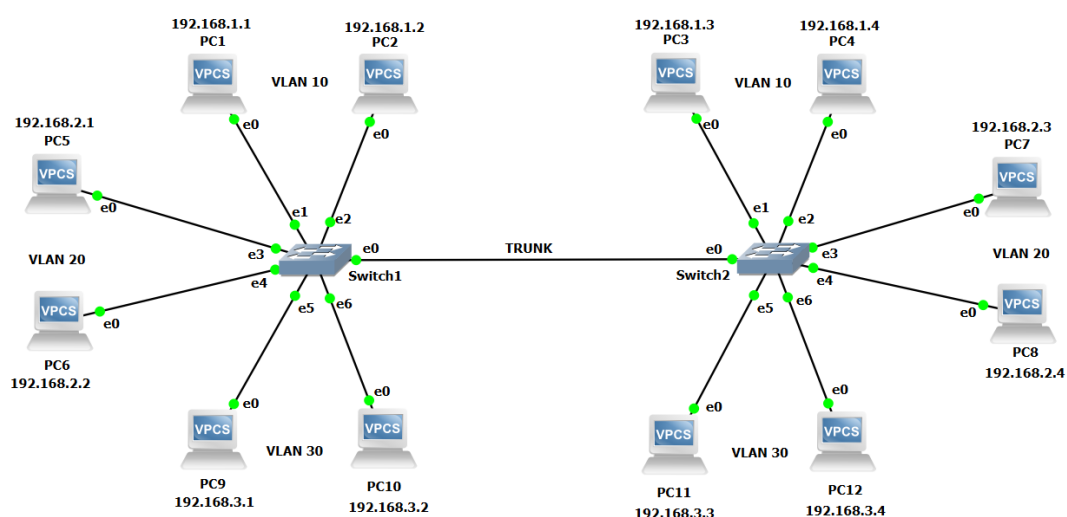


Figure 1: VLAN topology showing Switch1 and Switch2 connected by a trunk link

In GNS3

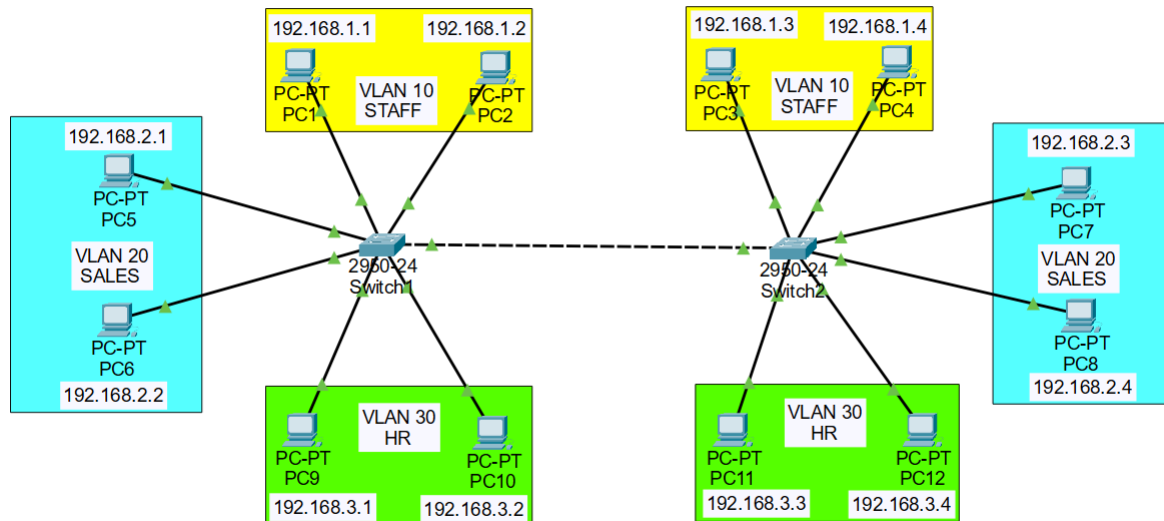


Figure 1.1: VLAN topology showing Switch1 and Switch2 connected by a trunk link in Cisco Packet Tracer

Switch Configuration

Cisco Packet Tracer Switch Configuration

The following steps were carried out on Switch1 through the CLI:

- Created VLAN 10 and named it STAFF
- Created VLAN 20 and named it SALES
- Created VLAN 30 and named it HR
- Set FastEthernet 0/24 as a trunk port to connect to Switch2
- Assigned FastEthernet 0/1 to 0/4 as access ports in VLAN 10
- Assigned FastEthernet 0/5 to 0/8 as access ports in VLAN 20
- Assigned FastEthernet 0/9 to 0/12 as access ports in VLAN 30

The same VLANs and port assignments were mirrored on Switch2 so that devices in matching VLANs could reach each other across the trunk link.

```

Switch>en
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name STAFF
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#int fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#int range fa0/1-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int range fa0/5-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#int range fa0/9-12
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write memory
Building configuration...
[OK]
Switch#exit

```

Figure 2: CLI commands used to create VLANs, configure the trunk port, and assign access ports on Switch1

GNS3 Switch Configuration

The same VLAN and port plan was recreated in GNS3. The screenshot below shows the port table for Switch1, with port 0 set as a trunk (dot1q) and the remaining ports set to access mode for VLANs 10, 20, and 30.

Switch1 configuration

General

Name: Switch1
Console type: none

Settings

Port: 0
VLAN: 1
Type: dot1q
QinQ EtherType: 0x8100

Add Delete

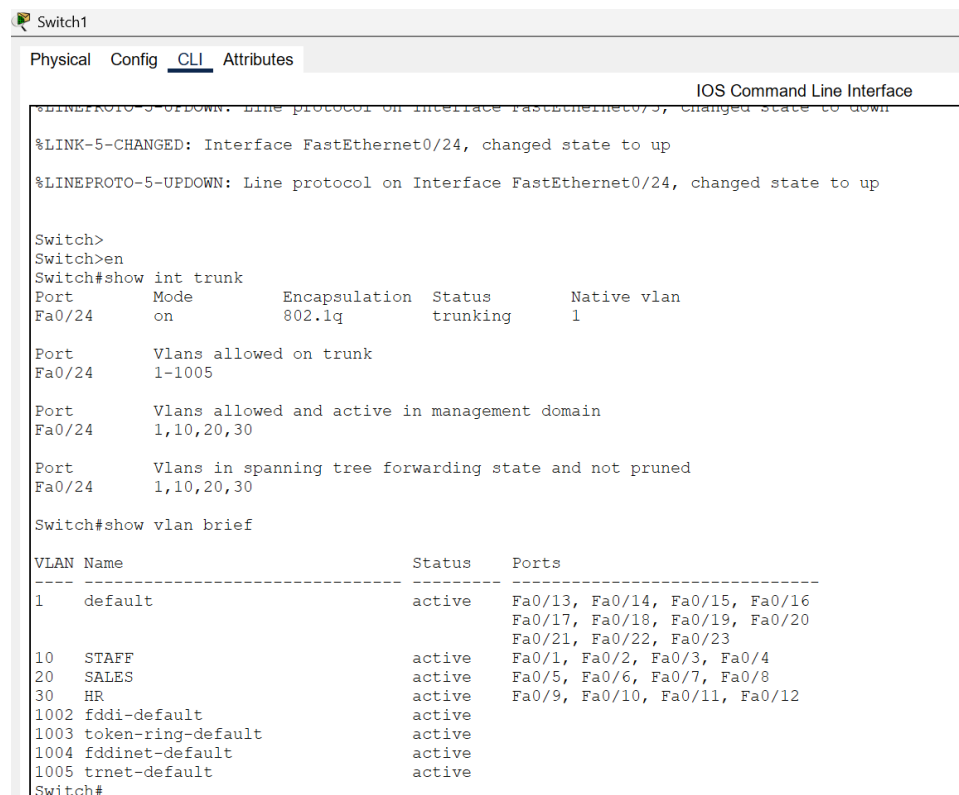
Ports

Port	VLAN	Type	EtherType
0	1	dot1q	
1	10	access	
2	10	access	
3	20	access	
4	20	access	
5	30	access	
6	30	access	
7	1	access	

Figure 3: GNS3 Switch1 configuration panel showing trunk and access port assignments

Verification

After the configuration was applied, the following commands were used on Switch1 to confirm the VLANs and trunk link were working correctly.



```
Switch1
Physical Config CLI Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch>
Switch>en
Switch#show int trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/24    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/24    1-1005

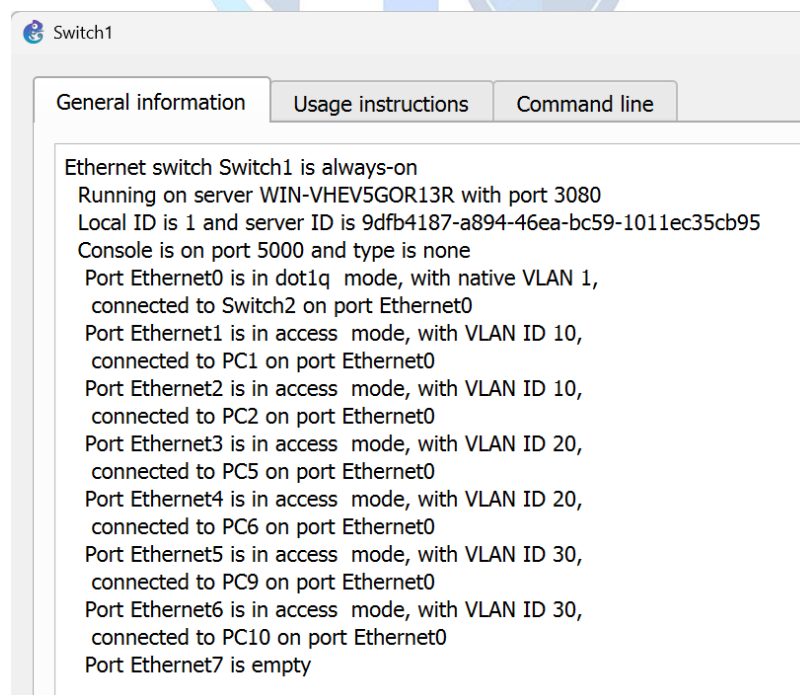
Port      Vlans allowed and active in management domain
Fa0/24    1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    1,10,20,30

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/13, Fa0/14, Fa0/15, Fa0/16,
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20,
                                           Fa0/21, Fa0/22, Fa0/23
10   STAFF                   active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
20   SALES                   active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
30   HR                      active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
1002 fddi-default           active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#
```

Figure 5: Output of show int trunk and show vlan brief on Switch1



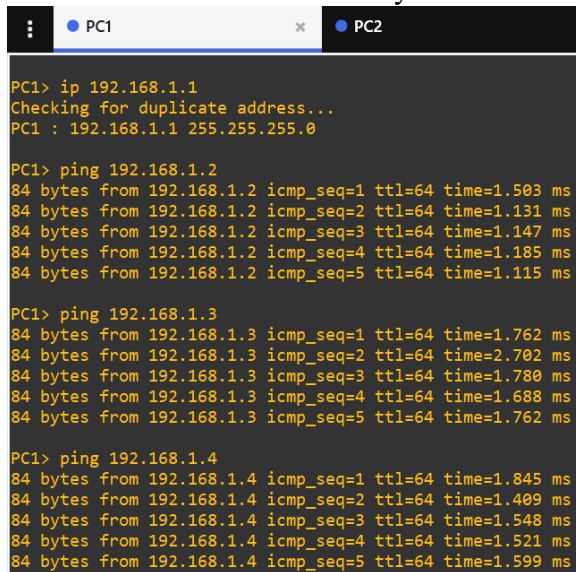
Ethernet switch Switch1 is always-on
Running on server WIN-VHEV5GOR13R with port 3080
Local ID is 1 and server ID is 9dfb4187-a894-46ea-bc59-1011ec35cb95
Console is on port 5000 and type is none
Port Ethernet0 is in dot1q mode, with native VLAN 1, connected to Switch2 on port Ethernet0
Port Ethernet1 is in access mode, with VLAN ID 10, connected to PC1 on port Ethernet0
Port Ethernet2 is in access mode, with VLAN ID 10, connected to PC2 on port Ethernet0
Port Ethernet3 is in access mode, with VLAN ID 20, connected to PC5 on port Ethernet0
Port Ethernet4 is in access mode, with VLAN ID 20, connected to PC6 on port Ethernet0
Port Ethernet5 is in access mode, with VLAN ID 30, connected to PC9 on port Ethernet0
Port Ethernet6 is in access mode, with VLAN ID 30, connected to PC10 on port Ethernet0
Port Ethernet7 is empty

Figure 4: GNS3 general information tab confirming each port's mode, VLAN ID, and connected device

The show int trunk output confirms that Fa0/24 is trunking using 802.1Q encapsulation, and that VLANs 1, 10, 20, and 30 are active and allowed on the trunk. The show vlan brief output confirms VLAN 10 (STAFF), VLAN 20 (SALES), and VLAN 30 (HR) are active, each with the correct ports assigned.

Ping Test Results

Ping tests were run between PCs in the same VLAN but on different switches, to confirm the trunk link correctly carries traffic for each VLAN.



```

PC1> ip 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.503 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.131 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.147 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=1.185 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.115 ms

PC1> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=1.762 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=2.702 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=1.780 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.688 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=1.762 ms

PC1> ping 192.168.1.4
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=1.845 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=1.409 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=1.548 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=1.521 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=1.599 ms
  
```

Figure 6: PC1 (VLAN 10, Switch1) successfully pinging PC2, PC3, and PC4, confirming VLAN 10 connectivity across both switches in GNS3

Fire	Last Status	Source	Destination	Type	Color	Time(sec)
	Successful	PC1	PC4	ICMP		0.000
	Successful	PC5	PC7	ICMP		0.000
	Successful	PC9	PC12	ICMP		0.000

Figure 7: Summary of successful ping tests between PC1-PC4 (VLAN 10), PC5-PC7 (VLAN 20), and PC9-PC12 (VLAN 30) across the trunk link in Cisco Packet Tracer

All ping tests were successful, confirming that devices in the same VLAN can communicate with each other regardless of which switch they are connected to.

Challenges Faced and Solutions

Challenge 1: Trunk configuration in Cisco Packet Tracer did not work at first. The issue was using the wrong cable type between the switches.

Solution: This was fixed by using the correct cable for the trunk connection.

Challenge 2: GNS3 does not provide a CLI or console for switches.

Solution: VLAN and port configuration in GNS3 had to be done manually through the switch UI instead of typing commands.

Key Learning Outcomes

- Learned how to create VLANs and assign names to them on a Cisco switch.
- Learned the difference between access ports and trunk ports, and when to use each.
- Learned how 802.1Q trunking allows multiple VLANs to travel over a single link between switches.
- Learned how to verify VLAN and trunk configuration using show vlan brief and show interfaces trunk.
- Confirmed VLAN segmentation works as intended using ping tests between same-VLAN devices on different switches.

Conclusion

This task gave hands on experience with VLAN configuration and network segmentation using both Cisco Packet Tracer and GNS3. Three VLANs (STAFF, SALES, and HR) were created and assigned to separate access ports on two switches, connected through a trunk link carrying all VLAN traffic between them.

Configuring the trunk in Packet Tracer required using the correct cable type, and GNS3 required manual UI based configuration since it has no CLI or console access for switches. Both challenges were resolved, and the VLAN setup was verified using show vlan brief and show interfaces trunk, along with ping tests between devices in the same VLAN across both switches.

All tests confirmed successful communication within each VLAN, showing a correctly working trunk link and proper VLAN segmentation across the network. This task strengthened understanding of how VLANs improve network security, organization, and traffic management, and built practical skill in configuring and verifying VLANs in enterprise style network environments.